

Rohit Acharya

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RESEARCH INTERESTS

Construction production planning and control · human judgement and review in automated monitoring systems · schedule performance and delay analysis · digital construction and progress measurement · computational modelling in civil engineering

EDUCATION

National Institute of Technology (NIT) Rourkela Odisha, India
Bachelor of Technology in Civil Engineering — CGPA: 7.96 / 10.0 Jan 2022 – May 2025

- **Thesis:** *Buckling Performance Evaluation of Functionally Graded Graphene-Reinforced Composite Plates under Thermal-Mechanical Loads*; Supervisor: Prof. Shishir Kumar Sahu.
- **Relevant coursework:** Construction Planning, Scheduling and Control; Estimating and Costing; Construction Materials; Probability and Statistics; Numerical Analysis; Finite Element Method; Programming and Data Structures.

Kathmandu Bernhardt Secondary School Kathmandu, Nepal
Higher Secondary Certificate, Science Stream — CGPA: 3.85 / 4.0 2018 – 2020

PROFESSIONAL EXPERIENCE

CloudFactory Lalitpur, Nepal
Construction Analyst — AI Construction Monitoring & Project Controls Platform Jul 2025 – Present

- **Production planning and control:** run line-of-balance reviews, pace-variation analysis, work-plan target tracking, and schedule status reporting for large commercial, industrial, and advanced-manufacturing capital projects; produce delay forecasts used by project teams to identify slippage.
- **Human review of automated results:** act as the review and escalation point where the platform's automated progress inference is uncertain or disputed; adjudicate ambiguous cases against captured evidence and record the reasoning.
- **Root-cause analysis for clients:** when a client disputes a reported result, trace it end to end through field capture, BIM and 3D model registration, activity configuration, dependency logic, automated inference, and human review, and report where the determination failed.
- **Progress measurement:** establish as-built status from 360° walkthrough imagery, drone survey, and 3D scans against BIM models and planned sequences, and quantify as-planned versus as-built deviation.
- **Standards and configuration:** turn recurring review failures into revised review standards, activity configuration, dependency rules, and auto-completion logic, reducing repeat issues across projects.

Envirocare Infrastructure Pvt. Ltd. (Shri Mahavir Ferro Alloys) Rourkela, India
Project Intern — Site Execution & Estimation Jan 2025 – May 2025

- Supervised civil execution works across industrial structures; carried out site inspections and prepared Bills of Materials and Running Account bills from field measurements and structural drawings.

Bureau of Indian Standards (BIS) Guwahati, India
Intern — Quality Assurance, Civil Engineering Division May 2024 – Jul 2024

- Applied Indian Standards to material compliance testing for cement, aggregates, and steel reinforcement; assisted in laboratory testing, document audits, and compliance report analysis.

RESEARCH EXPERIENCE

Undergraduate Researcher — Computational Modelling & Simulation Jun 2024 – Dec 2024
Department of Civil Engineering, NIT Rourkela Supervisor: Prof. Shishir Kumar Sahu Rourkela, India

- Built finite element models in **ABAQUS** to study buckling, post-buckling, and free vibration of functionally graded composite plates under coupled thermal-mechanical loading.
- Designed and ran a parametric study across 12+ input variables; processed the resulting data in **Python** and **MATLAB**, and checked every trend against three published studies before accepting it.

Seminar Researcher — FRP Strengthening of Reinforced Concrete 2024
Department of Civil Engineering, NIT Rourkela CE4900: Seminar & Technical Writing Rourkela, India

- Reviewed CFRP, GFRP, and BFRP strengthening systems for aging reinforced concrete and derived analytical formulations for capacity gains, later encoded into a deployed design tool.

PROJECTS

Structural Health Monitoring Dashboard | [Live](#) | Next.js, TypeScript, Recharts 2025 – Present

- Deployed a monitoring prototype ingesting simulated infrastructure sensor streams, with rolling z-score anomaly detection and a WebSocket path for live instrument data.

FG-GRC Plate Buckling & Vibration Simulator | [Live](#) | React, Python, FastAPI, MATLAB 2025

- Turned the thesis method into an interactive tool implementing micromechanics and plate-theory eigenvalue solutions, with a Python/FastAPI backend.

FRP-RC Section Analyzer | [Live](#) | React, TypeScript, Tailwind 2025

- Dual-code (IS 456 / ACI 440.1R) section analysis aid computing capacity, moment–curvature response, and deflection, with each calculation step shown.

TECHNICAL SKILLS

Planning & Controls: line of balance, pace-variation analysis, delay forecasting, work-plan target tracking, as-planned versus as-built deviation, schedule status reporting, root-cause investigation.

Programming: Python (NumPy, SciPy, Pandas, Matplotlib), MATLAB, FastAPI, TypeScript, JavaScript, React, Next.js; three deployed engineering applications.

Data & AI: time-series analytics, anomaly detection, parametric study design, failure analysis of automated inference systems, ML foundations (Python for Data Science, AI & Development, IBM).

Modelling: ABAQUS 2023, SAP2000, ETABS, STAAD.Pro; finite element, buckling, and modal analysis.

Digital Construction: BIM-model comparison, 360° imagery, drone survey and 3D scan interpretation, AutoCAD, Revit, BIM.

Languages: Nepali (native), English (full professional proficiency), Hindi (fluent).

AWARDS & HONORS

Aspire Institute Leadership Program Scholar, 2024 - competitive global leadership program.

District Topper, Baglung District, Nepal, 2018 - among the highest-performing students in the district in the national Secondary Education Examination, GPA 3.95 / 4.0.

National Finalist, National Astronomy Olympiad (NAO), Nepal, 2019 & 2020.

LEADERSHIP & SERVICE

Class Representative, Department of Civil Engineering, NIT Rourkela (2022–2025): elected across multiple semesters; acted as the point of contact between the batch and the faculty, raised and resolved student concerns, and coordinated academic scheduling and departmental communication

Student Mentor, Institute Counselling Service, NIT Rourkela (2023–2025): provided academic and personal guidance to junior students under the Institute's peer-mentoring programme

Coordinator, Higher Education Committee, NIT Rourkela (2024–2025): led the committee connecting students to internship and fellowship opportunities across NIT Rourkela and partner institutions, and organised webinars and seminars on graduate study and research pathways

Founder & Lead Educator, Learners Club, Kathmandu (2020–2024): founded a free STEM education platform so school students across Nepal could keep learning during COVID-19; Physics, Chemistry, and Mathematics content reaching 20,000+ views ([channel](#))

Student Member, American Society of Civil Engineers (2021–2024): General Member

CERTIFICATIONS

Python for Data Science, AI & Development — IBM / Coursera, 2024 | Foundations: Data, Data, Everywhere — Google / Coursera, 2024